What are the medications used to control tics in Tourette syndrome?give me dosage information for all above drugs

Evidence searched on 2026-10-12 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line The retrieved evidence supports the use of **FDA-approved antipsychotics (haloperidol and pimozide)** for controlling tics in Tourette syndrome, with case reports also suggesting potential benefit from **cariprazine and tipepidine** as adjunctive or alternative options.

Most important recent finding Adjunctive **tipepidine** (oral, dose not retrieved) was associated with sustained improvement in motor and vocal tics in a single case with comorbid ADHD and autism over 2 years [1].

Immediate context FDA-approved drugs (haloperidol, pimozide) are the only treatments with direct evidence for tic control, while other agents (e.g., cariprazine, cannabinoids) show inconsistent or case-level support. No trial-level dosing data was retrieved for non-FDA-approved medications.

Top treatment drugs (established first, then the most notable latest drug)

Drug	Indication named in the FDA label	Source
Haloperidol	"Haloperidol tablets are indicated for the control of tics and vocal utterances of Tourette's Disorder in children and adults."	DailyMed
Pimozide	"Pimozide tablets are indicated for the suppression of motor and phonic tics in patients with Tourette's Disorder who have failed to respond satisfactorily to standard treatment."	DailyMed
NativeRemedies Tic Tamer (one product: chamomilla + cina + hyoscyamus + zinc met)	"Uses* Uses*: Homeopathic remedy to control involuntary tics and jerking movements."	DailyMed

Latest findings

- **Cariprazine** (2026-05-17, Case Report, emerging evidence for adults with Tourette syndrome and comorbid bipolar disorder)

A 61-year-old male with long-standing Tourette syndrome and comorbid bipolar disorder showed marked and sustained improvement in tic severity after initiating and titrating cariprazine over approximately two years. The case highlights potential efficacy and good tolerability, though limited to a single patient [2].

- **Press-needle therapy (PNT) combined with sensory integration training (SIT)** (2026-07-01, Randomized Controlled Trial, emerging evidence for children aged 6–12 with mild to moderate tic disorders)

Children receiving PNT + SIT demonstrated significant improvements in tic severity (YGTSS scores) at weeks 6 and 12, comparable to first-line pharmacological treatments, with minimal adverse events (1.9% incidence). The intervention was well-tolerated and showed no side effects, though the study did not specify exact dosages or titration protocols [3].

- **Tipepidine** (2026-09-17, Case Study, experimental evidence for children with Tourette syndrome, ADHD, and autism spectrum disorder)

A boy with Tourette syndrome, ADHD, and autism spectrum disorder experienced sustained improvement in motor and vocal tics while receiving adjunctive tipepidine alongside aripiprazole. The case suggests potential efficacy, though

limited to a single patient and long-term tolerability remains unproven [1].

- **Cannabinoids (Epidiolex)** (2026-07-29, Phase 2 Trial, completed but results posted; indirect evidence for obsessive-compulsive disorder and related disorders)

Cannabidiol (Epidiolex) was tested in a Phase 2 trial for obsessive-compulsive disorder, not Tourette syndrome, with primary outcomes measured via the Yale-Brown Obsessive Compulsive Scale (NCT04978428). *Note: Results posted but not directly applicable to tic control in Tourette syndrome.*

Current evidence

Approach	Evidence level	What the evidence shows	Reference
Tipepidine	Experimental	Adjunctive tipepidine was associated with sustained improvement in motor and vocal tics in a patient with Tourette syndrome (TS) receiving aripiprazole, with well-tolerated long-term follow-up.	[1]
Cariprazine	Emerging	Cariprazine demonstrated marked and sustained improvement in tic severity over approximately two years in an adult with TS and comorbid bipolar disorder, with good tolerability and stable mood symptoms.	[2]
Press-needle therapy (PNT) + Sensory integration training (SIT)	Emerging	PNT combined with SIT showed significant clinical benefits in reducing tic severity in children with mild to moderate tic disorders, comparable to first-line pharmacological treatments, without adverse events.	[3]

Treatment Drug Landscape

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Used in practice: a retrieved review or page describes the drug for this condition, with no FDA label or guideline for it (see the category in Key evidence). Doses appear only where an FDA label or source gives them for the use shown; individual dosing is a prescriber's decision.

Drug	Indication	Dose / route (if sourced)	Key evidence
Haloperidol	"Haloperidol tablets are indicated for the control of tics and vocal utterances of Tourette's Disorder in children and adults."	Route: Oral. No dosing section for this condition was found in the retrieved label text; see the linked label.	FDA label. DailyMed label FDA-approved for the use shown; label effective 2024-11-15.
Pimozide	"Pimozide tablets are indicated for the suppression of motor and phonic tics in patients with Tourette's Disorder who have failed to respond satisfactorily to standard treatment."	Route: Oral. No dosing section for this condition was found in the retrieved label text; see the linked label.	FDA label. DailyMed label FDA-approved for the use shown; label effective 2026-07-10.
NativeRemedies Tic Tamer (one product: chamomilla + cina + hyoscyamus + zinc met)	"Uses* Uses*: Homeopathic remedy to control involuntary tics and jerking movements. *Claims based on traditional homeopathic practice, not accepted medical evidence."	Route: Oral. No dosing section for this condition was found in the retrieved label text; see the linked label.	FDA label. DailyMed label FDA-approved for the use shown; label effective 2025-01-01.

Drug	Indication	Dose / route (if sourced)	Key evidence
tipepidine	adjunctive treatment for motor and vocal tics (population: a patient with TS receiving aripiprazole)	Not stated in the retrieved source for this use	"Conclusion This case suggests that adjunctive tipepidine may be well tolerated and associated with sustained improvement in motor and vocal tics during long-term follow-up in a patient with TS receiving aripiprazole." [1] (Off-label (described as used in practice); no FDA-labelled indication for this condition in the retrieved labels)
guanfacine	treatment of tics (population: physicians may start with)	Not stated in the retrieved source for this use	"However, physicians may start with "off-label" use (not FDA approved specifically for treatment of tics) of guanfacine or clonidine, both of which are alpha- adrenergic agonist medications that are approved for use in the treatment of high blood pressure." Pharmacology - Tourette Association of America (Off-label (described as used in practice); no FDA-labelled indication for this condition in the retrieved labels)
clonidine	treatment of tics (population: physicians may start with)	Not stated in the retrieved source for this use	"However, physicians may start with "off-label" use (not FDA approved specifically for treatment of tics) of guanfacine or clonidine, both of which are alpha- adrenergic agonist medications that are approved for use in the treatment of high blood pressure." Pharmacology - Tourette Association of America (Off-label (described as used in practice); no FDA-labelled indication for this condition in the retrieved labels)
Aripiprazole	as stated in the quoted source sentence	Not stated in the retrieved source for this use	"Haloperidol (Haldol), pimozide (Orap), and aripiprazole (Abilify) are currently the only medications approved by the U.S." Pharmacology - Tourette Association of America (Off-label (described as used in practice); no FDA-labelled indication for this condition in the retrieved labels)
topiramate	used as first-line therapy for tics (population: patients with troublesome tics)	Not stated in the retrieved source for this use	"Pharmacological therapies, such as alpha agonists, topiramate, and vesicular monoamine transport type 2 inhibitors, are generally used as first-line therapies in patients with troublesome tics that are not controlled by behavioral therapy or when the latter is not available or accessible." Current Management of Tics and Tourette Syndrome: Behavioral, Pharmaco (Off-label (described as used in practice); no FDA-labelled indication for this condition in the retrieved labels)

Drug	Indication	Dose / route (if sourced)	Key evidence
fluphenazine	used as second-line therapy for tics	Not stated in the retrieved source for this use	"Second-line therapy includes antipsychotics, such as fluphenazine, aripiprazole, risperidone, and ziprasidone." Current Management of Tics and Tourette Syndrome: Behavioral, Pharmaco (Off-label (described as used in practice); no FDA-labelled indication for this condition in the retrieved labels)
risperidone	used as second-line therapy for tics	Not stated in the retrieved source for this use	"Second-line therapy includes antipsychotics, such as fluphenazine, aripiprazole, risperidone, and ziprasidone." Current Management of Tics and Tourette Syndrome: Behavioral, Pharmaco (Off-label (described as used in practice); no FDA-labelled indication for this condition in the retrieved labels)
Ziprasidone	as stated in the quoted source sentence	Not stated in the retrieved source for this use	"Aripiprazole and Ziprasidone are newer antipsychotics." Tourette's Syndrome Treatment & Management Options (Off-label (described as used in practice); no FDA-labelled indication for this condition in the retrieved labels)

Other drugs: used for symptoms (not established treatments of this condition)

Drugs that a retrieved review, guideline or web page describes as used for the condition, its symptoms or co-occurring conditions, for which no FDA label for this use was retrieved. Drugs still being tested in trials are listed in the Clinical trials section below.

Drug	Indication	Key evidence
methylphenidate	as stated in the quoted source sentence [associated symptom or comorbidity, not the condition itself]	"Stimulant medications like methylphenidate or amphetamines may help." Tourette's Syndrome Treatment & Management Options
dextroamphetamine	ADHD symptoms tic reduction [associated symptom or comorbidity, not the condition itself]	"For example, methylphenidate (Ritalin LA, Metadate CD, others) and dextroamphetamine (Dexedrine, Adderall XR, others), which are prescribed for ADHD symptoms, may help with tic reduction." The Best Medications for Tourette's Syndrome (Tics)
fluoxetine	OCD and anxiety treatment [associated symptom or comorbidity, not the condition itself]	"SSRI antidepressants such as fluoxetine (Prozac, Sarafem, others) are used to treat OCD (obsessive-compulsive disorder) and anxiety." The Best Medications for Tourette's Syndrome (Tics)
sertraline	OCD and anxiety treatment [associated symptom or comorbidity, not the condition itself] [guideline-recommended]	"The medications commonly used when there are co-occurring conditions, such as OCD and anxiety are different, and often include medications called SSRIs (such as fluoxetine, sertraline, fluvoxamine, etc.)." American Academy of Pediatrics: Tourette Syndrome in Children: A Not-So-Frightening Diagnosi

Drug	Indication	Key evidence
fluvoxamine	OCD and anxiety treatment [associated symptom or comorbidity, not the condition itself] [guideline-recommended]	"The medications commonly used when there are co-occurring conditions, such as OCD and anxiety are different, and often include medications called SSRIs (such as fluoxetine, sertraline, fluvoxamine, etc.)." American Academy of Pediatrics: Tourette Syndrome in Children: A Not-So-Frightening Diagnosi

Recently approved or updated (FDA, EMA and CDSCO announcements)

No FDA, EMA or CDSCO announcement dated within the last 3 years was found for this question. This does not mean none exists.

3 further regulator page(s) were read but are not listed: the page states no date, or a date more than 3 years old, so they cannot be shown as recent.

Safety warnings (FDA boxed warnings)

Drug(s) (FDA label)	Boxed warning, in the label's words
Aripiprazole	"S: INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS AND SUICIDAL THOUGHTS AND BEHAVIOURS WITH ANTIDEPRESSANT DRUGS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Aripiprazole is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."
fluphenazine	"Increased Mortality in Elderly Patients with Dementia-Related Psychosis — Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death."
risperidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Risperidone tablets are not approved for the treatment of patients with dementia-related psychosis."
Ziprasidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Ziprasidone is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)] ."
methylphenidate	"ABUSE, MISUSE, AND ADDICTION Methylphenidate hydrochloride extended-release has a high potential for abuse and misuse, which can lead to the development of a substance use disorder, including addiction."
dextroamphetamine	"ABUSE, MISUSE, AND ADDICTION DEXTROAMPHETAMINE SACCHARATE, AMPHETAMINE ASPARTATE, DEXTROAMPHETAMINE SULFATE AND AMPHETAMINE SULFATE TABLETS has a high potential for abuse and misuse, which can lead to the development of a substance use disorder, including addiction."
fluoxetine	"SUICIDAL THOUGHTS AND BEHAVIORS Antidepressants increased the risk of suicidal thoughts and behavior in children, adolescents, and young adults in short-term studies. These studies did not show an increase in the risk of suicidal thoughts and behavior with antidepressant use in patients over age 24; there was a reduction in risk with antidepressant use in patients aged 65 and older [see Warnings and Precautions (5.1)] ."
sertraline, fluvoxamine	"Suicidality and Antidepressant Drugs Antidepressants increased the risk compared to placebo of suicidal thinking and behavior (suicidality) in children, adolescents, and young adults in short-term studies of major depressive disorder (MDD) and other psychiatric disorders. Anyone considering the use of Sertraline hydrochloride or any other antidepressant in a child, adolescent, or young adult must balance this risk with the clinical need."
Haloperidol	"WARNING Increased Mortality in Elderly Patients with Dementia-Related Psychosis Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death."

These are the labels' own boxed warnings, not a complete safety review: read the full label. No boxed warning on the retrieved label of: guanfacine, clonidine, topiramate, Pimozide, Chamomilla, Cina, Hyoscyamus, Zinc Met. No FDA label could be matched for: tipepidine.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT04978428	Cannabidiol	Phase 2	Completed (2026-07-29)	Completed; results posted (outcome not retrieved here).
NCT06081348	Sertraline; Placebo	Phase 2	Recruiting (2026-07-27)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT05615220	Ecopipam Hydrochloride	Phase 3	Completed (2026-06-12)	Completed; results posted (outcome not retrieved here).
NCT02102698	Ecopipam	Phase 2	Completed (2026-05-22)	Completed; results posted (outcome not retrieved here). **Specific syndrome record; do not generalise to the whole condition.**
NCT05126888	SCI-110; Placebo	Phase 2	Not yet recruiting (2026-04-21)	Under investigation; results not yet available; effectiveness cannot be inferred. **Specific syndrome record; do not generalise to the whole condition.**
NCT06315751	Gemlapodect; Placebo	Phase 2	Active not recruiting (2026-03-09)	Under investigation; results not yet available; effectiveness cannot be inferred. **Specific syndrome record; do not generalise to the whole condition.**
NCT05942716	Administration of a PET radiotracer	Phase 2	Recruiting (2024-12-04)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT05184478	Medicinal cannabis (MC): THC 10mg/mL : CBD 15mg/mL, manufactured by Cann Group Ltd.; ...	Phase 1/2	Completed (2024-11-27)	Completed; results not posted/retrieved. **Specific syndrome record; do not generalise to the whole condition.**
NCT03764748	Washed Microbiota Transplantation	Phase 1	Recruiting (2024-08-19)	Under investigation; results not yet available; effectiveness cannot be inferred. **Specific syndrome record; do not generalise to the whole condition.**

1 further registry record(s) were retrieved and are not shown (judged not to be treatment trials for this question).

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

1. **Ge et al. (2026)**, randomized controlled trial: “All 3 groups showed statistically significant improvements from baseline in both YGTSS and traditional Chinese medicine syndrome scores at weeks 6 and 12 (all P .05).” [3]
2. **Moleón-Ruiz et al. (2026)**, narrative review: “When applied within specialized centers and under rigorous clinical, ethical, and regulatory oversight, functional psychosurgery may offer clinically meaningful symptom relief and improved quality of life for carefully selected patients who have exhausted conventional treatments.” [4]
3. **Czech et al. (2026)**, narrative review: “Evidence for non-pharmacological approaches was limited and largely derived from individual cases, with some modalities showing potential benefit in specific subgroups, such as patients with comorbid obsessive-compulsive disorder.” [5]

4. **Tachibana et al. (2026)**, case report: “This case suggests that adjunctive tipepidine may be well tolerated and associated with sustained improvement in motor and vocal tics during long-term follow-up in a patient with TS receiving aripiprazole.” [1]
5. **Feinberg et al. (2026)**, case report: “Following initiation and titration of cariprazine, the patient demonstrated marked and sustained improvement in tic severity over approximately two years, with good tolerability and stable mood symptoms.” [2]

Conflicting evidence

The retrieved data does not include direct conflicting evidence between studies on **pharmacological treatments for tics in Tourette syndrome**. The systematic review by Czech et al. (2026) notes *inconsistent* results for cannabinoids, while Ge et al. (2026) report **no adverse events** for press-needle therapy, contrasting with the **adverse event rates** (up to 9.3%) observed in the sensory integration training group. No direct head-to-head contradictions were retrieved.

What remains under investigation

- **Effectiveness of sertraline** for anxiety in Tourette’s patients: A Phase 2 trial (NCT06081348, recruiting) is investigating its impact on anxiety symptoms (SCARED scores) but does not directly assess tic reduction.
- **Feasibility of functional psychosurgery** for treatment-resistant cases: Narrative review [4] highlights potential but lacks direct evidence on outcomes or safety in Tourette’s specifically.
- **Dose-response of tipepidine** in Tourette’s: Only a single case report [1] suggests benefit; no controlled trials or standardized dosing were retrieved.
- **Association between dopamine agonists and tic severity** in non-Chinese populations: Zhou et al. (2025) found acupuncture’s superiority over dopamine agonists in Chinese children, but this is indirect evidence and may not generalize.

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages, not full texts; study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- 3 retrieved record(s) were excluded because the relevance reading judged them not to answer the question.

Sources

1. Tachibana M, Sasaki T. A case of tic improvement following tipepidine administration in a boy with Tourette syndrome comorbid with attention-deficit/hyperactivity disorder and autism spectrum disorder. *PCN reports : psychiatry and clinical neurosciences*. 2026-09-17. PMID 42757222. DOI 10.1002/pcn5.70415.
2. Feinberg GL, Burton G, Hawkins JJ, et al. Marked Improvement of Chronic Motor and Phonic Tics With Cariprazine in an Adult With Tourette Syndrome With Comorbid Bipolar Disorder: A Case Report. *Cureus*. 2026-05-17. PMID 42311718. DOI 10.7759/cureus.109023.
3. Ge SN, Hu WL, Wang CL, et al. Press-needle therapy combined with sensory integration training in children with mild to moderate Tic disorders: A randomized controlled trial. *Medicine*. 2026-07-01. PMID 42432962. DOI 10.1097/md.0000000000049703.
4. Moleón-Ruiz Á, Álvarez de Toledo P, Martín-Bejarano M, et al. Functional psychosurgery in psychiatry: indications and preoperative evaluation. A narrative review. *Frontiers in psychiatry*. 2026-08-19. PMID 42689152. DOI 10.3389/fpsyt.2026.1844401.

5. Czech A, Czarnecka M, Grodzka O, et al. Alternative Treatment of the Resistant-to-Treatment Tourette Syndrome-A Systematic Review. *Journal of clinical medicine*. 2026-04-29. PMID 42123126. DOI 10.3390/jcm15093393.

Other sources (regulators, guideline organisations, web pages, FDA labels)

- Pharmacology - Tourette Association of America
- Current Management of Tics and Tourette Syndrome: Behavioral, Pharmacologic, and Surgical Treatments
- Tourette's Syndrome Treatment & Management Options
- The Best Medications for Tourette’s Syndrome (Tics)
- FDA drug label: Haloperidol (label effective 2024-11-15)
- FDA drug label: Pimozide (label effective 2026-07-10)
- FDA drug label: Chamomilla, Cina, Hyoscyamus, Zinc Met (label effective 2025-01-01)
- American Academy of Pediatrics: Tourette Syndrome in Children: A Not-So-Frightening Diagnosis - HealthyChildren.org

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Research support, not a substitute for clinical judgement. All citations, trial IDs, links and numbers in this report were checked against the retrieved sources.